ILARIA PULITO

PhD Student in Biomedical Engineering | Predoctoral Researcher

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PhD Student in Biomedical Engineering at UPM and Predoctoral Researcher at CSIC Neural Engineering Laboratory. Specializing in neurotechnology, wearable systems, and closed-loop neuromodulation, with expertise in physiological signal processing (HD-EMG, heart sounds) and clinical deep learning.

RESEARCH & ACADEMIC EXPERIENCE

PhD Student & Predoctoral Researcher
12/2025 – Present

Universidad Politécnica de Madrid (UPM) & Cajal Neuroscience Center (CSIC)

Neural Engineering Laboratory | Supervisors: Filipe Oliveira Barroso; co-director: Álvaro Gutiérrez

Research Area: Neurotechnology, Wearable Systems, Closed-Loop Neuromodulation

- Development of a wearable device dedicated to non-invasive tremor suppression in clinical applications.
- Processing and analysis of High-Density Electromyography (HD-EMG) signals for neural activity decoding.
- Implementation of closed-loop stimulation strategies based on real-time biosignal monitoring.

Research Trainee – Biomedical Engineering
09/2024 – 02/2025

INESC TEC – Institute for Systems and Computer Engineering, Technology and Science

Porto, Portugal | Bioengineering applied to non-invasive diagnostics

- Developed and optimized deep learning algorithms and advanced signal processing techniques for automated heart sound analysis.
- Focused on non-invasive early detection and clinical diagnostic support for pulmonary hypertension.

Didactic Tutor – Physics I
03/2021 – 07/2024

Politecnico di Torino

Turin, Italy

- Conducted weekly tutoring sessions and guided exercises for first-year engineering students in Physics I.
- Supported students in mastering classical physics concepts, problem-solving, and exam preparation.

EDUCATION & TRAINING

Master's Degree in Biomedical Engineering
09/2022 – 03/2025

Politecnico di Torino – Department of Mechanical and Aerospace Engineering

Specialization: Biomedical Instrumentation | 120 CFU

Thesis: *Pulmonary Hypertension Detection via Deep Learning of Heart Sounds*

Erasmus+ Research Mobility
09/2024 – 02/2025

INESC TEC / Universidade do Porto

Porto, Portugal | Focus: Bioengineering & Signal Processing

Bachelor's Degree in Biomedical Engineering
09/2019 – 10/2022

Politecnico di Torino

Industrial Engineering (L-9) | 180 CFU

PUBLICATIONS

Innocente, C., Boemio, M., Lorenzetti, G., **Pulito, I.**, Romagnoli, D., Saponaro, V., Marullo, G., Ulrich, L., & Vezzetti, E. (2025). Deep Learning-Based Lip-Reading for Vocal Impaired Patient Rehabilitation. *[VENUE PLACEHOLDER]*. Available at: ResearchGate [LINK PLACEHOLDER]

SKILLS

Languages & Libraries:	Python, MATLAB, PyTorch, TensorFlow, Keras, scikit-learn, SolidWorks, LaTeX
Biomedical & Signals:	Physiological Signal Processing (HD-EMG, Heart Sounds), Wearable Devices, Closed-Loop Neuromodulation, Clinical Deep Learning
Languages (CEFR):	Italian (Native), English (C1), Spanish (B1), Portuguese (A2)

PUBLICATIONS

Innocente, C., Boemio, M., Lorenzetti, G., **Pulito, I.**, Romagnoli, D., Saponaro, V., Marullo, G., Ulrich, L., & Vezzetti, E. (2025). Deep Learning-Based Lip-Reading for Vocal Impaired Patient Rehabilitation. *[VENUE PLACEHOLDER]*. Available at: ResearchGate